

Research-Style Analysis of Step 2 CK Error Intake Dataset

Dataset scope: approximately two weeks of longitudinal Step 2-style question intake capturing confidence state, diagnosis, discriminating clue, error type, and management reasoning.

Abstract

This report analyzes an early longitudinal dataset derived from a structured Step 2 CK error-intake system. The dataset demonstrates a highly coherent cognitive-error architecture rather than diffuse underperformance. The dominant failure mode was management sequencing and algorithm selection, followed by diagnostic confusion between closely related entities. Internal medicine domains — particularly cardiology, nephrology, pulmonary/critical care, infectious disease, and endocrinology — represented the largest concentration of misses. Confidence-state analysis revealed that most misses occurred in the medium-confidence range, suggesting a large zone of recoverable performance through refinement rather than broad content acquisition.

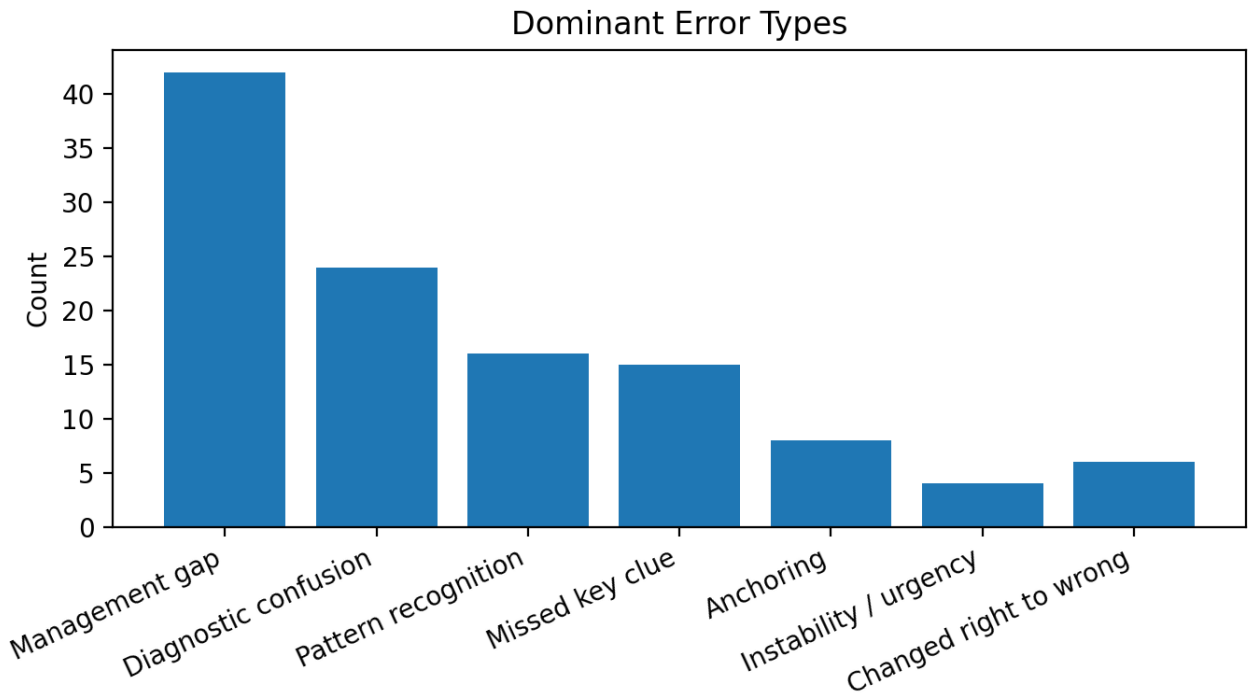
Dataset Structure

The intake system captured the following variables for each missed or educational question: confidence state, distractor elimination performance, diagnosis, discriminating clue, error type, management insight, and structured cognitive takeaway. This level of granularity substantially exceeds conventional incorrect-question review logs.

Primary Findings

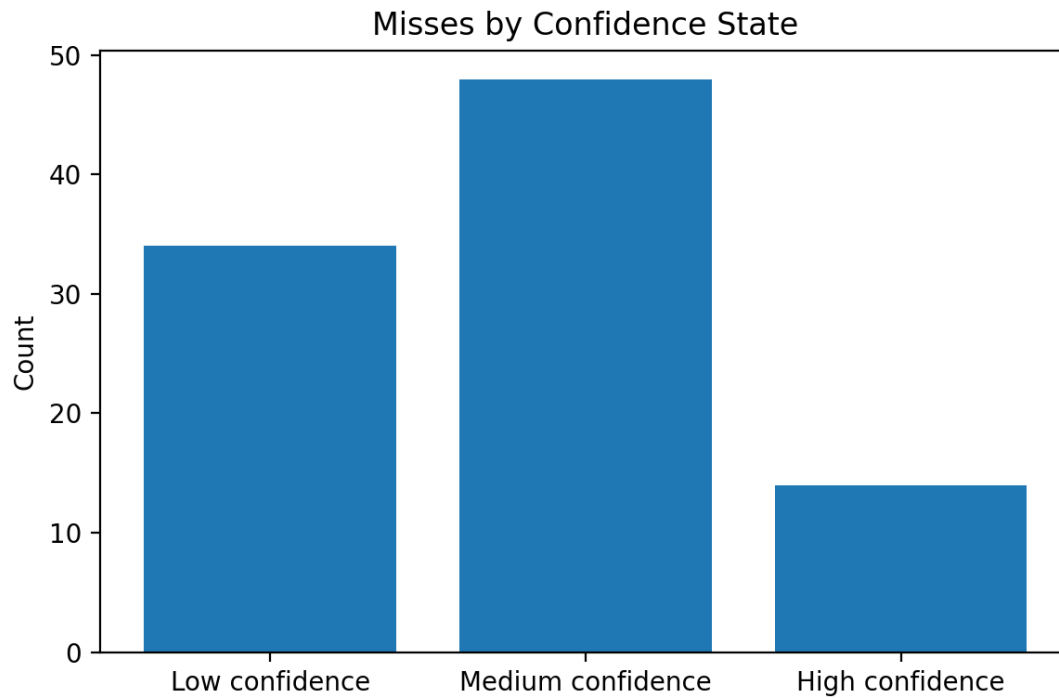
1. Management Sequencing is the Dominant Bottleneck

The single most common failure mode was `management_gap`. These misses were not random factual failures; they clustered around selecting the wrong step in a known algorithm. Examples included stabilization sequencing, contraindications, escalation thresholds, confirmatory testing, and disposition logic.



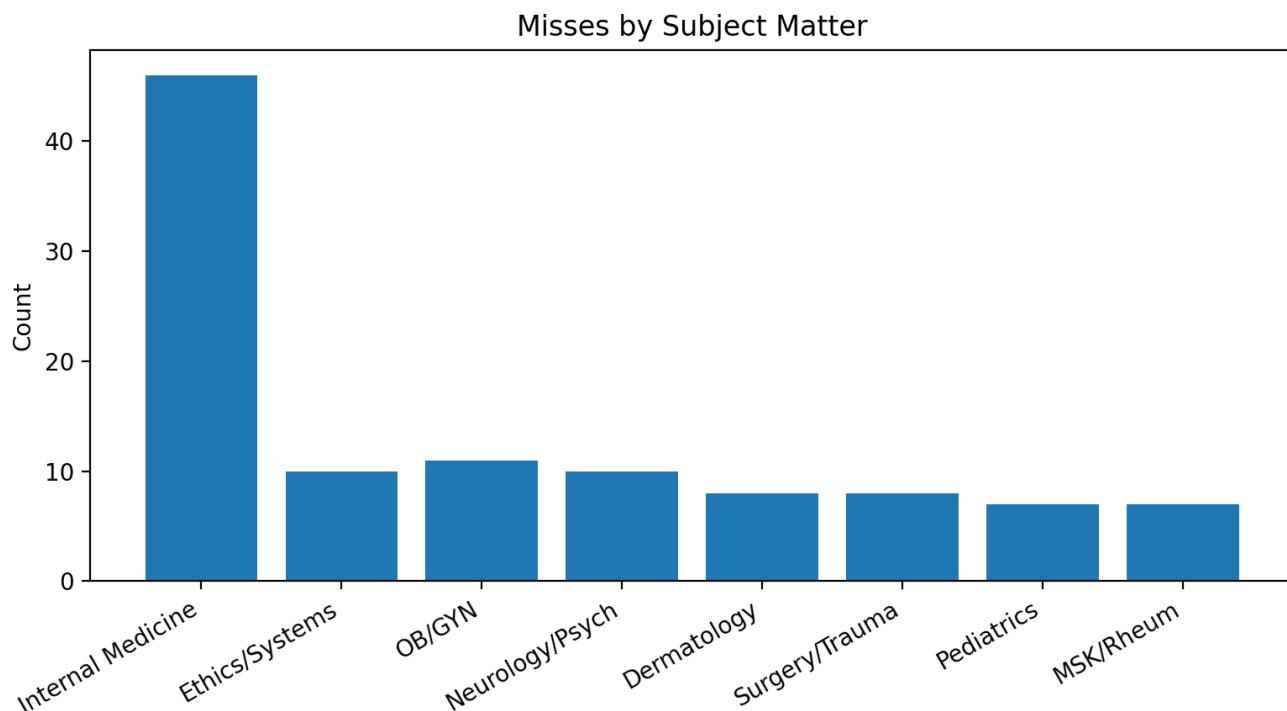
2. Medium-Confidence Errors are the Highest-Yield Zone

Most misses occurred in the medium-confidence state. This pattern implies that the examinee frequently recognizes the syndrome family correctly but fails at precise differentiation or exact next-step execution. This is a fundamentally different cognitive profile from diffuse ignorance.



3. Subject-Matter Concentration

The error distribution was strongly concentrated within Internal Medicine domains. Renal, cardiopulmonary, endocrine, and infectious disease pathways dominated the miss profile. Secondary clusters occurred in OB/GYN, neurology, ethics/systems, dermatology, and trauma.



Longitudinal Interpretation

Although the dataset spans only approximately two weeks, there is evidence that the miss profile is evolving from coarse recognition failures toward more sophisticated algorithmic and discrimination-based misses. Early misses more commonly reflected anchoring and incomplete pattern recognition. Later misses increasingly involved nuanced management ladders, contraindications, escalation timing, and discrimination between closely related diagnoses.

Confidence Calibration

High-confidence misses represented a smaller fraction of total errors but carried disproportionate educational value. These questions frequently involved premature closure, anchoring, or overgeneralization from familiar schemas. Examples included aortic pathology, pulmonary pathology, neurologic localization, and toxicology.

Interpretive Discussion

The most important finding is that the error landscape is highly structured. The dataset does not support the interpretation of broad foundational deficiency. Instead, the performance profile resembles an examinee transitioning from recognition-level reasoning toward algorithmic clinical reasoning. The principal deficits involve execution precision rather than conceptual emptiness.

This distinction matters. A student with diffuse foundational weakness requires broad content acquisition. By contrast, this dataset suggests a learner who increasingly identifies

the correct disease space but loses points on: (1) sequencing, (2) instability recognition, (3) contraindication logic, (4) discriminating clues, and (5) management escalation thresholds.

Strategic Recommendations

- Prioritize management ladders over broad rereading.
- Create contrast-style cards emphasizing discriminating clues between near-neighbor diagnoses.
- Use dedicated anti-anchoring review for high-confidence misses.
- Separate review workflows by confidence state.
- Continue structured intake exactly as currently designed; the ontology is unusually high quality for educational analytics.

Conclusion

The intake architecture demonstrates unusually high analytic maturity for a medical education workflow. The system already resembles a lightweight clinical reasoning knowledge graph rather than a traditional incorrect-question log. The emerging profile suggests substantial Step 2 performance headroom through targeted refinement of management algorithms, confidence calibration, and discriminating-clue recognition.